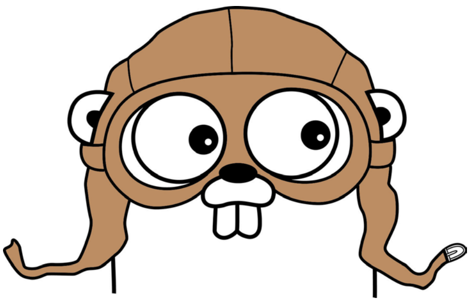
The Best Machine Learning Libraries in Golang

Golang has a number of stable and well supported open source machine learning libraries. As it allows for faster production-ready development and a large number of well-supported repositories, it has got glowing endorsements from the open source community. Golang is an emerging open source programming language for machine learning aspirants. In this article, we will discuss the topmost libraries in Golang.



Golang was created by Google in 2007 with the aim of designing an efficient, compiled programming language. It has numerous advantages in terms of reliability and production:

- Build time is very fast in comparison with other languages
- Run time performance is excellent
- Excellent concurrency support
- Rich set of libraries, particularly for machine learning
- Forced error handling to minimise unforeseen exceptions
- Great adaption environment as projects grow
- Great dependency management
- Simpler IDE and debugging
- Native support

Some of the world's most successful technology companies use Golang as the main language of their production systems and actively contribute to its development. Cloudflare uses Golang in many in-house software projects as well as parts of bigger vital projects. Google itself uses Golang to solve its problems. Uber uses Golang extensively for high throughput and low latency requirements as well as non-disruptive background loading. Dailymotion is a global video streaming service company with a network of over 250 million people that uses Golang extensively. It has developed an application called Asteroid using Golang to manage its Wireguard server. The application improved the efficiency of the company, while adding and removing access to its infrastructure. Golang is emerging as a powerful open source machine learning tool due to its great features and libraries.

Machine learning libraries in Golang

GoLearn

GoLearn is the most vital package for Golang. It can be used for many machine learning algorithms. Density based spatial clustering (DBSCAN), random forest (RF), k-nearest neighbors (KNN), Naïve Bayes (NB), neural network (NN) and principal component analysis (PCA) are the main machine learning algorithms of this package. In order to install this package, you will need to have a compatible compiler installed. After installing Go and the system dependencies, run:

```
go get -t -u -v github.com/sjwhitworth/golearn
```

Run the following to complete the installation:

```
cd $GOPATH/src/github.com/sjwhitworth/golearn
go get -t -u -v ./...
```

Gorgonia

This is a lower level library. The architecture for the model has to be built by the developer. The main striking feature of this library is its ability to deal with multi-dimensional arrays easily and efficiently. The other feature is performance. Since it is a lower level library, building a model takes more steps. However, all the steps are easy. In order to perform any application using Gorgonia, we first need to import some libraries:

```
package main
Import {
            "fmt"
            "log"
            . "gorgonia.org/gorgonian"
                  }
```